for loop :-

```
int i;
for (i=0 ; i<n ; i++)
{
    printf("Hello");
}
```

Output :-

Hello
Hello
Hello

```
for ( i=10 ; i>5 ; i--);
{
    printf("Hello");
}
```

1 time

```
3) for (i=1 ; i<=7 ; i--)
{
    printf("Hello");
}
```

i
1 <=7

1)

```
for (i=1 ; i<5 ; i++)
{
    printf("Hello");
}
```

i n
1 5
2 3 4 5

1 2 3 4 — 4 times

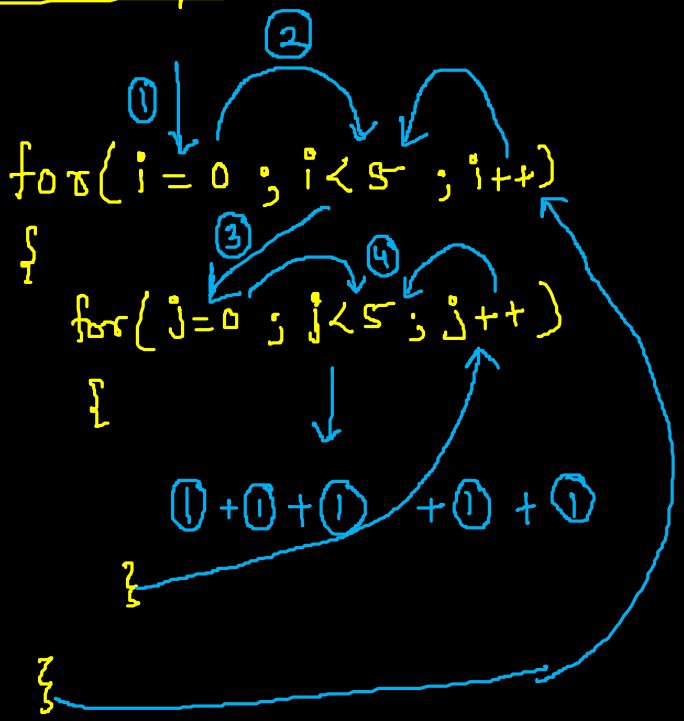
2)

```
for(i=0 ; i<=5 ; i++)
{
    printf ("Hello");
}
```

i n
0 5
1 2 3 4 5 6

1 2 3 4 5 6 — 6 times

Nested for loop:-



i	j	
~~0~~	~~0~~	0
1	~~1~~	
	~~2~~	
	~~3~~	
	~~4~~	
	5	

i = 0 j [0-4] = 5

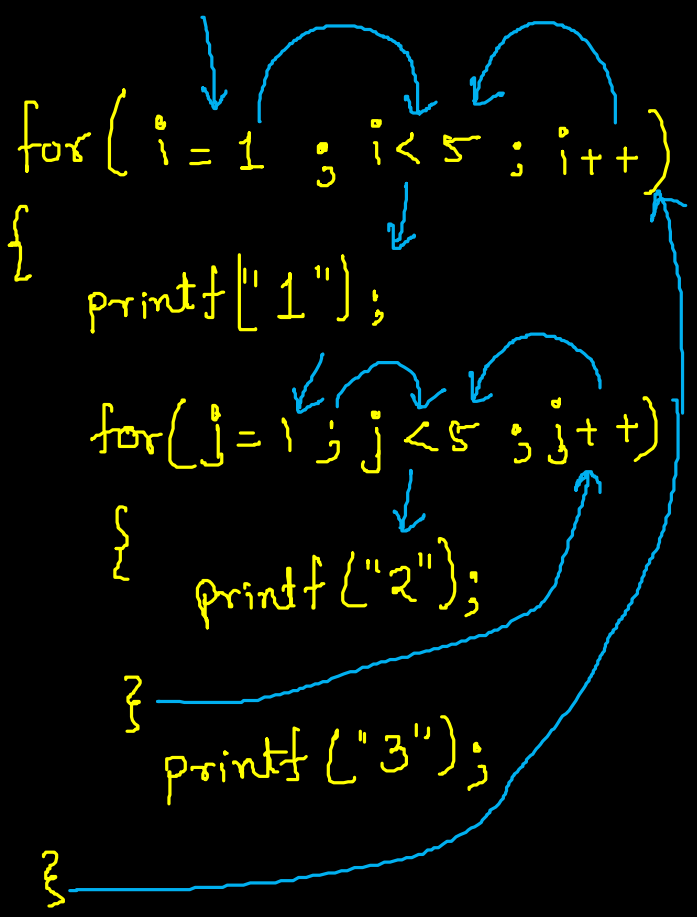
i = 1 j [0-4] = 5

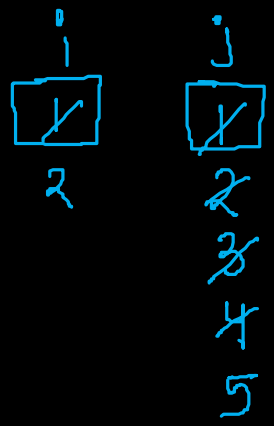
⋮

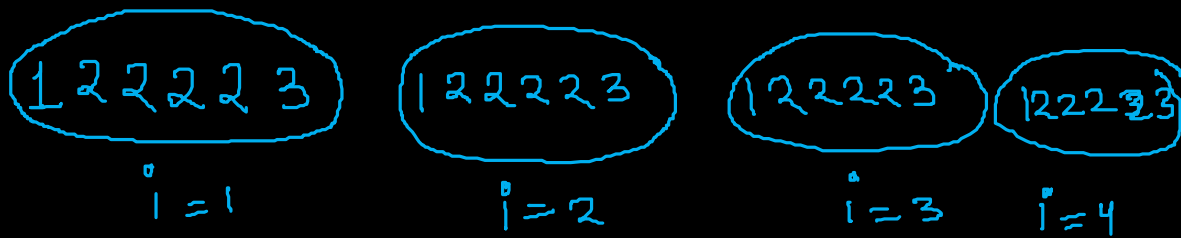
i = 4 j [0-4] = 5

i = 5 Break

25


for (i = 1 ; i < 5 ; i++)
{
printf("1");
for(j = 1 ; j < 5 ; j++)
{
printf("2");
}
printf("3");
}


i
j
1
1
2
2
3
4
5


122223
122223
122223
122223
i=1
i=2
i=3
i=4

```
for ( i=0 ; i<5 ; i++)
{
    printf(" 1");
    for (j=0 ; j<5 ; j++)
    {
        printf ("2");
        for (k=0 ; k<5 ; k++)
        {
            printf ("3");
        }
        printf (" 4");
    }
    printf (" 5 ");
}
```

i = [0 - 4] = 5 times

printf(" 1"); → (5 times)

j = [0-4] = 5 times

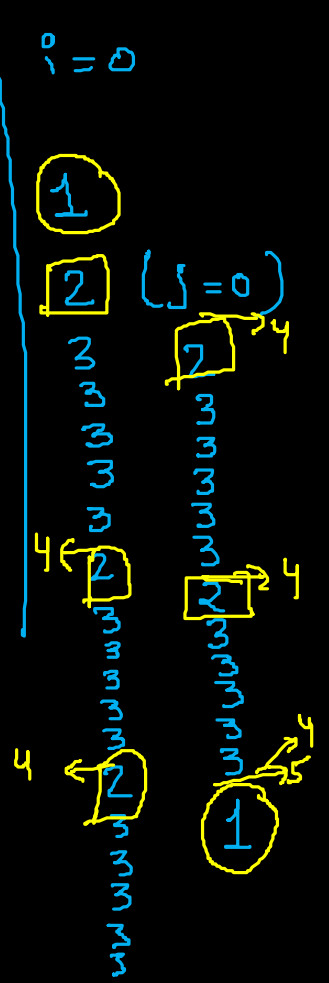
printf ("2"); → 5 × 5 = (25 times)

k = [0-4] = 5 times

printf ("3"); → 5 × 5 × 5 = (125 times)

printf (" 4"); → 25 times

Printf(" 5 "); → 5 times


i=0
1
2
(j=0
2
4
3
3
3

```
for(i=0;i<5;i++)
{
    printf("1");
}
for(j=0;j<5;j++)
{
    printf("2");
}
```

5 times + 5 times

11111 22222 → 10 times

1-D → single for loop →

2-D

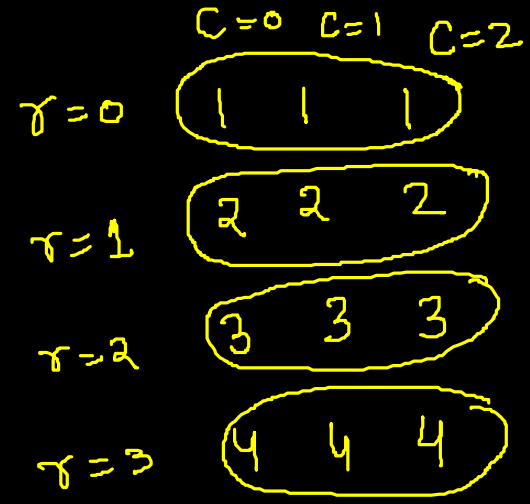
```
for(r=0;r<4;r++)          → 5
{                           x
    for(c=0;c<3;c++)      → 5
    {
        printf("1");      → 25
    }
}
```

	c=0	1	2
r=0	1	1	1
1	2	2	2
2	3	3	3
3	4	4	4

no of col = 3

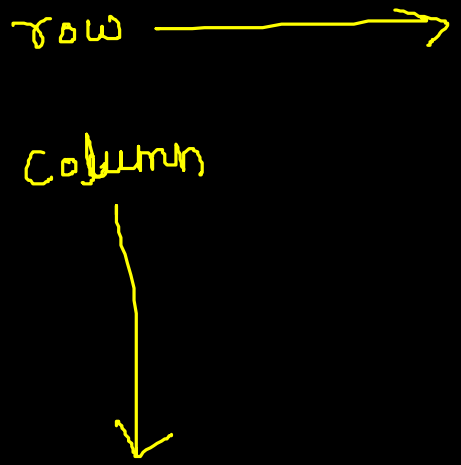
no. of row = 4

no. of cells = 12

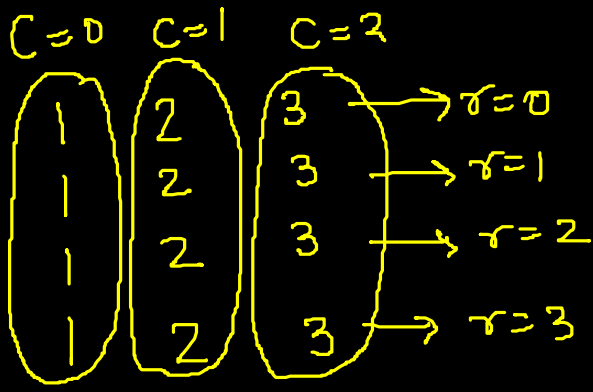


row major

12



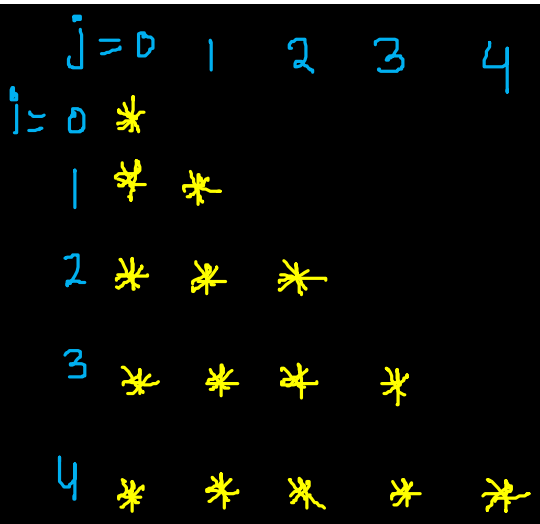

```
for(r=0; r<4; r++)
{
    for(c=0; c<3; c++)
        printf("%d", r+1);
    printf("\n");
}
```



12

column major

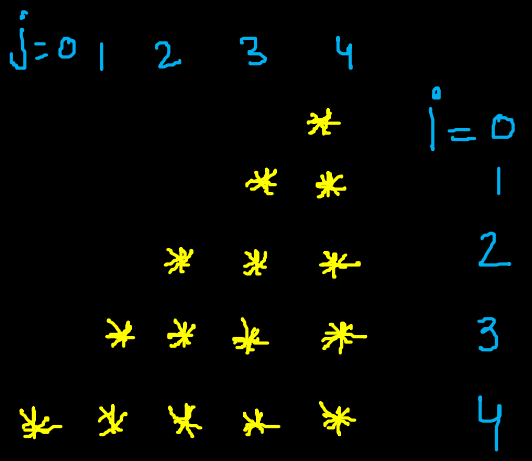
```
for(c=0; c<3; c++)
{
    for(r=0; r<4; r++)
        printf("%d", r+1);
    printf("\n");
}
```



$j = [0 \ldots\ldots i] \Rightarrow j <= i$

```
for(i=0 ; i<=4 ; i++)
{
    for(j=0 ; j<=i ; j++)
    {
        printf("*");
    }
    printf("\n");
}
```

i = 0 — j = 0
i = 1 — j = 0, 1
i = 2 — j = 0, 1, 2
i = 3 — j = 0, 1, 2, 3
i = 4 — j = 0, 1, 2, 3, 4




```
for( i=0 ; i<=4 ;i++)
{
  for ( j=0 ; j<=4 ; j++)
  {
    if ( j >= 4-i)
      printf(" * ");
    else
      printf("   ");
  }
  printf(" \n");
}
```

i = 0 j = 4

i = 1 j = 3, 4

i = 2 j = 2, 3, 4

i = 3 j = 1, 2, 3, 4

i = 4 j = 0, 1, 2, 3, 4

(4-0) = 4

(4-1) = 3

(4-2) = 2

(4-3) = 1

(4-4) = 0

(j >= 4-i)

break , Continue

```
for( i = 0 ; i < 10 ; i++)
{
    if( i == 4)
        break;
}
```

→ it will break the loop inside which it is present

i = 0, 1, 2, 3, 4

```
for( i = 0 ; i < 10 ; i++)
{
    if (i == 4)
    continue ;
    printf (" Hello");
}
```

i = 0, 1, 2, 3, 4, 5, 6, 7, 8, 9

4 → skip

9 times